

Inorganic Chemistry Practical: Topic 10 – Group 2 & Gravimetric Analysis

10 Most Frequently Asked Practical Examination Questions (Past 4 Years Analysis: 2021–2024)

CANDIDATE NAME	URWAH	CENTRE NUMBER	PK082
COMPONENT	Paper 3 – Advanced Practical Skills (Past 10 Years Archive)	CANDIDATE NUMBER	0142
TIME ALLOWED	2 Hours 30 Minutes	TOTAL MARKS	100 Marks

INSTRUCTIONS TO CANDIDATES:

- Answer all questions in the spaces provided on the question paper.
- Give working for all calculations and record all experimental readings to the required precision:
 - Burette readings to the nearest 0.05 cm³; balance readings to 0.01 g / 0.001 g.
 - Thermometer readings to the nearest 0.5 °C.
- You may use an HB pencil for any rough working, graphs, or apparatus diagrams.
- Non-exact numerical answers should be given to three significant figures unless stated otherwise.
- You will need a scientific calculator and the Qualitative Analysis Notes provided on pages at the back.

LABORATORY INVESTIGATION OVERVIEW:

This laboratory pack contains authentic Cambridge Advanced Practical examinations. Each practical scenario provides complete procedures, chemical specifications (FA codes), hazard warnings, structured data collection tables, graphical plotting grids, and comprehensive multi-step mathematical and observational deductions.

Question 1: Determination of Water of Crystallization in Hydrated Magnesium Sulfate to Constant Mass

Cambridge Syllabus Reference: 9701/31/M/J/24/Q2 · [10 Marks]

Method: 3.25 g of $\text{MgSO}_4 \cdot x\text{H}_2\text{O}$ is heated in a crucible with lid. The sample is heated gently for 3 minutes, strongly for 7 minutes, cooled in a desiccator, and weighed. Reheating yields constant residue mass = 1.59 g.

(a) Complete the crucible weighing table below with recorded masses. [3]

Measurement Description	Mass / g
Mass of empty crucible + lid	
Mass of crucible + lid + hydrated sample before heating	
Mass of crucible + lid + sample after 1st heating	
Mass of crucible + lid + sample after 2nd heating (to constant mass)	
Mass of anhydrous salt residue	
Mass of water of crystallization lost	

(b) Calculate the mass of anhydrous MgSO_4 remaining and the mass of water driven off. [2]

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(c) Calculate moles of MgSO_4 and H_2O , and deduce the integer value of x . [Mr: $\text{MgSO}_4 = 120.4$, $\text{H}_2\text{O} = 18.0$] [3]

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(d) Explain the role of the desiccator in this experiment. [2]

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Question 2: Thermal Decomposition of Basic Copper(II) Carbonate (Malachite)**Cambridge Syllabus Reference: 9701/33/M/J/23/Q2 · [10 Marks]**

Reaction: $\text{CuCO}_3 \cdot \text{Cu}(\text{OH})_2(\text{s}) \rightarrow 2\text{CuO}(\text{s}) + \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{g})$

Method: 4.42 g of green malachite powder is heated in a boiling tube connected to a limewater trap. The powder turns jet black and gas bubbles through the limewater. Final black residue mass = 3.18 g.

(a) Calculate the theoretical percentage mass loss for pure $\text{CuCO}_3 \cdot \text{Cu}(\text{OH})_2$. [$M_r = 221.0$, $(\text{CO}_2 + \text{H}_2\text{O}) = 62.0$] [3]

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(b) Calculate the experimental mass loss and the percentage purity of the malachite sample. [3]

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(c) State the observation in the limewater and write an ionic equation for this reaction. [2]

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(d) Explain why the delivery tube must be removed from limewater before removing the heat source. [2]

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Question 3: Gravimetric Analysis of Calcium Carbonate Thermal Decomposition**Cambridge Syllabus Reference: 9701/34/M/J/23/Q2 · [10 Marks]**Reaction: $\text{CaCO}_3(\text{s}) \rightarrow \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$

Data: 2.50 g of limestone is strongly heated in a porcelain crucible until no further mass loss is observed.

Residue mass = 1.51 g. [Mr: CaCO_3 = 100.1, CaO = 56.1, CO_2 = 44.0]

(a) Calculate the mass of carbon dioxide lost during heating. [2]

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(b) Calculate the moles of CO_2 evolved and hence the mass of CaCO_3 decomposed. [3]

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(c) Calculate the percentage purity of the limestone sample. [2]

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(d) Explain why a Bunsen burner flame alone cannot fully decompose barium carbonate, BaCO_3 . [3]

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Question 4: Comparative Thermal Decomposition of Group 2 Nitrates: Calcium Nitrate vs Barium Nitrate

Cambridge Syllabus Reference: 9701/35/M/J/22/Q2 · [10 Marks]

Equation: $2\text{M}(\text{NO}_3)_2(\text{s}) \rightarrow 2\text{MO}(\text{s}) + 4\text{NO}_2(\text{g}) + \text{O}_2(\text{g})$

Samples of $\text{Ca}(\text{NO}_3)_2$ and $\text{Ba}(\text{NO}_3)_2$ are heated in boiling tubes in a fume cupboard. Brown fumes are observed and a glowing splint is tested at the mouth of each tube.

(a) Identify the gas responsible for the brown fumes and the gas that relights a glowing splint. [2]

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(b) State which nitrate decomposes more readily (requires less heating) and explain why. [4]

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(c) Write the balanced chemical equation for the thermal decomposition of calcium nitrate. [2]

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(d) State the hazard associated with NO_2 gas and specify the required safety measure. [2]

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Question 5: Qualitative Investigation of Solubility Trends in Group 2 Hydroxides**Cambridge Syllabus Reference: 9701/32/M/J/24/Q3 · [10 Marks]**

Aqueous solutions of Mg^{2+} , Ca^{2+} , and Ba^{2+} (0.1 mol dm^{-3}) are tested with 2.0 mol dm^{-3} NaOH dropwise and in excess.

- Mg^{2+} : Thick white precipitate insoluble in excess.
- Ca^{2+} : Faint white precipitate formed.
- Ba^{2+} : Solution remains clear (no precipitate).

(a) Deduce the trend in solubility of Group 2 hydroxides down the group from Mg to Ba. [2]

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(b) Explain this trend in terms of lattice energy and hydration enthalpy changes down Group 2. [4]

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(c) Write the ionic equation for the precipitation of magnesium hydroxide. [2]

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(d) State one common pharmaceutical use of a suspension of magnesium hydroxide. [2]

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Question 6: Qualitative Investigation of Solubility Trends in Group 2 Sulfates**Cambridge Syllabus Reference: 9701/33/O/N/21/Q3 · [10 Marks]**

Aqueous solutions of Mg^{2+} , Ca^{2+} , and Ba^{2+} are tested with dilute sulfuric acid, $\text{H}_2\text{SO}_4(\text{aq})$.

- Mg^{2+} : No precipitate (clear solution).
- Ca^{2+} : Faint/slight white precipitate.
- Ba^{2+} : Dense, heavy white precipitate immediately formed.

(a) Deduce the trend in solubility of Group 2 sulfates down the group. [2]

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(b) Explain why sulfate solubility decreases down the group whereas hydroxide solubility increases. [4]

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(c) Write the ionic equation for the precipitation of barium sulfate. [2]

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(d) Explain why barium sulfate can be safely swallowed by patients undergoing X-ray imaging despite Ba^{2+} being highly toxic. [2]

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Question 7: Analysis of Experimental Technique: Crucible Lid Positioning in Thermal Analysis

Cambridge Syllabus Reference: 9701/31/M/J/23/Q2 · [10 Marks]

In thermal gravimetric analysis, instructions specify:

1. Heat with the crucible lid partially ajar.
2. Cool with the crucible lid completely closed in a desiccator.

(a) Explain the consequence if the lid is left completely closed during strong heating. [3]

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(b) Explain the consequence if the crucible has no lid at all during heating. [3]

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(c) Explain why the lid must be placed tightly shut during the cooling phase. [2]

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(d) State why tongs must always be used rather than fingers when moving crucibles. [2]

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Question 8: Mitigation of Mechanical Spitting and Thermal Shock in Gravimetric Preparations

Cambridge Syllabus Reference: 9701/35/M/J/24/Q2 · [10 Marks]

When hydrated crystals are placed directly over a roaring Bunsen flame, violent cracking sounds and spitting of solid particles occur. A student records an apparent formula of $\text{MgSO}_4 \cdot 9\text{H}_2\text{O}$ instead of $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$.

(a) Explain how solid spitting leads to an erroneously high calculated value for x . [3]

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(b) Describe the proper heating sequence that prevents mechanical spitting of crystals. [3]

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(c) Explain why placing a hot crucible directly onto a cold stone bench can cause thermal shock. [2]

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(d) Identify the proper heat-resistant support used to hold a crucible on a tripod. [2]

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Question 9: Chemical Confirmation of Transition Metal Oxidation State in Decomposition Residue

Cambridge Syllabus Reference: 9701/34/O/N/23/Q2 · [10 Marks]

Following the thermal decomposition of basic copper carbonate, a black residue remains in the tube. A student must prove chemically that the black powder is copper(II) oxide, CuO.

(a) Name the acid used to dissolve the black residue and state the observation. [2]

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(b) Describe the two-step test using aqueous ammonia to confirm copper(II) ions in the blue solution. [4]

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(c) Write the ionic formula of the complex ion responsible for the deep royal blue color. [2]

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(d) Write the balanced chemical equation for the reaction of CuO with sulfuric acid. [2]

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Question 10: Physical Principles of Suck-Back Prevention and Laboratory Safety in Gas Evolution

Cambridge Syllabus Reference: 9701/32/O/N/23/Q2 · [10 Marks]

When heating solids in a boiling tube with an attached delivery tube submerged in limewater, extinguishing the flame before removing the delivery tube causes liquid to rush up the tube.

(a) Explain the physical mechanism that drives suck-back when the flame is removed. [4]

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(b) Explain why suck-back of cold liquid into a hot glass tube is dangerous. [2]

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(c) State two methods to prevent suck-back during an experiment. [2]

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(d) State the correct angle for clamping a boiling tube during strong heating of solids. [2]

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Examiner Mark Scheme & Indicative Observation Guide

Confidential Marking Points & Error Tolerances for Inorganic Chemistry Practical: Topic 10 – Group 2 & Gravimetric Analysis

Question 1: Determination of Water of Crystallization in Hydrated Magnesium Sulfate to Constant Mass [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	Table formatted correctly with appropriate units (/ g) [1]; balance readings to 2 d.p. [1]; constant mass agreement within ± 0.02 g [1].	[3]
(b)	Mass of MgSO_4 = 1.59 g [1]; Mass of water = 3.25 - 1.59 = 1.66 g [1].	[2]
(c)	Moles MgSO_4 = 1.59 / 120.4 = 0.01321 mol [1]; Moles H_2O = 1.66 / 18.0 = 0.09222 mol [1]; $x = 0.09222 / 0.01321 = 6.98 \sim 7$ ($\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$) [1].	[3]
(d)	Allows the hot crucible to cool to room temperature in an environment free of moisture, preventing anhydrous MgSO_4 from reabsorbing water vapor from air [2].	[2]

Question 2: Thermal Decomposition of Basic Copper(II) Carbonate (Malachite) [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	Theoretical % loss = $(62.0 / 221.0) \times 100 = 28.05\%$ [3].	[3]
(b)	Experimental mass loss = 4.42 - 3.18 = 1.24 g [1]; % purity = $(1.24 / (0.2805 \times 4.42)) \times 100 = 100\%$ (or exact proportional purity calculation) [2].	[3]
(c)	Limewater turns milky / forms white precipitate [1]; $\text{Ca}^{2+}(\text{aq}) + 2\text{OH}^-(\text{aq}) + \text{CO}_2(\text{g}) \rightarrow \text{CaCO}_3(\text{s}) + \text{H}_2\text{O}(\text{l})$ [1].	[2]
(d)	To prevent cold limewater being sucked back into the hot tube as cooling gas contracts, which would shatter the hot glass tube [2].	[2]

Question 3: Gravimetric Analysis of Calcium Carbonate Thermal Decomposition [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	Mass of CO_2 = 2.50 - 1.51 = 0.99 g [2].	[2]
(b)	Moles CO_2 = 0.99 / 44.0 = 0.0225 mol [1]; Moles CaCO_3 = 0.0225 mol [1]; Mass CaCO_3 = 0.0225 x 100.1 = 2.252 g [1].	[3]
(c)	% purity = $(2.252 / 2.50) \times 100 = 90.1\%$ [2].	[2]
(d)	Ba^{2+} has a larger ionic radius and lower charge density than Ca^{2+} , causing significantly less polarization of the carbonate ion; hence BaCO_3 has a much higher decomposition temperature exceeding standard Bunsen flame temperature [3].	[3]

Question 4: Comparative Thermal Decomposition of Group 2 Nitrates: Calcium Nitrate vs Barium Nitrate [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	Brown fumes: Nitrogen dioxide, NO_2 [1]; Relights glowing splint: Oxygen, O_2 [1].	[2]
(b)	Calcium nitrate decomposes more readily [1]; Ca^{2+} is smaller than Ba^{2+} and has a higher charge density [1]; Ca^{2+} polarizes the nitrate anion more strongly [1], weakening the N-O covalent bond and lowering thermal stability [1].	[4]
(c)	$2\text{Ca}(\text{NO}_3)_2(\text{s}) \rightarrow 2\text{CaO}(\text{s}) + 4\text{NO}_2(\text{g}) + \text{O}_2(\text{g})$ [2].	[2]
(d)	NO_2 is toxic and corrosive to respiratory system [1]; must be carried out in a fume cupboard [1].	[2]

Question 5: Qualitative Investigation of Solubility Trends in Group 2 Hydroxides [10 Marks]

Part	Marking Points & Expected Observations	Marks
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(a)	Solubility of Group 2 hydroxides increases down the group from $\text{Mg}(\text{OH})_2$ (least soluble) to $\text{Ba}(\text{OH})_2$ (most soluble) [2].	[2]
(b)	Down the group, cation radius increases while OH^- radius is small and fixed [1]; lattice energy decreases faster than the hydration enthalpy of the cation [2]; hence enthalpy of solution becomes more exothermic / less endothermic, increasing solubility [1].	[4]
(c)	$\text{Mg}^{2+}(\text{aq}) + 2\text{OH}^-(\text{aq}) \rightarrow \text{Mg}(\text{OH})_2(\text{s})$ [2].	[2]
(d)	Used as an antacid (Milk of Magnesia) to neutralize excess stomach hydrochloric acid without causing caustic burns [2].	[2]

Question 6: Qualitative Investigation of Solubility Trends in Group 2 Sulfates [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	Solubility of Group 2 sulfates decreases down the group from MgSO_4 (soluble) to BaSO_4 (insoluble) [2].	[2]
(b)	Sulfate ion (SO_4^{2-}) is very large compared to OH^- [1]; down the group, lattice energy changes very little because anion size dominates [1]; however, hydration enthalpy of the cation decreases significantly as cation radius increases [1]; hence $\Delta H_{\text{solution}}$ becomes increasingly endothermic, decreasing solubility [1].	[4]
(c)	$\text{Ba}^{2+}(\text{aq}) + \text{SO}_4^{2-}(\text{aq}) \rightarrow \text{BaSO}_4(\text{s})$ [2].	[2]
(d)	BaSO_4 has an extremely low solubility product (virtually insoluble in water and stomach acid), so negligible free toxic Ba^{2+} ions enter the bloodstream [2].	[2]

Question 7: Analysis of Experimental Technique: Crucible Lid Positioning in Thermal Analysis [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	Water vapor / steam cannot escape freely, building up pressure which can blow off the lid or condense on the underside and drip back into the salt residue [3].	[3]
(b)	Rapid boiling and thermal turbulence can cause solid salt crystals to 'spit' out of the crucible, resulting in an erroneously high recorded mass loss [3].	[3]
(c)	Prevents the anhydrous hygroscopic residue from absorbing moisture from the air while cooling [2].	[2]
(d)	Avoids severe burns from hot porcelain and prevents oils, sweat, and moisture from fingertips transferring onto the crucible and altering measured mass [2].	[2]

Question 8: Mitigation of Mechanical Spitting and Thermal Shock in Gravimetric Preparations [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	Lost solid crystals reduce the final mass of residue [1]; the calculated mass loss is assumed to be water vapor only [1]; this overestimates moles of water relative to residue, yielding $x > 7$ [1].	[3]
(b)	Begin by heating gently with a low, waving yellow/blue flame for 2-3 minutes to allow water to vaporize smoothly [1.5]; only after sputtering ceases, heat strongly with a roaring non-luminous flame [1.5].	[3]
(c)	Rapid, uneven thermal contraction of the porcelain causes internal mechanical stress that cracks the crucible [2].	[2]
(d)	Pipe-clay triangle (or silica triangle) [2].	[2]

Question 9: Chemical Confirmation of Transition Metal Oxidation State in Decomposition Residue [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	Dilute sulfuric acid, H_2SO_4 [1]; the black solid dissolves to form a clear sky-blue solution of $\text{CuSO}_4(\text{aq})$ [1].	[2]
(b)	Step 1: Add dilute NH_3 dropwise: a pale blue precipitate of $\text{Cu}(\text{OH})_2$ forms [2]; Step 2: Add excess NH_3 : the precipitate dissolves to give an intense deep royal blue solution of $[\text{Cu}(\text{NH}_3)_4(\text{H}_2\text{O})_2]^{2+}$ [2].	[4]
(c)	$[\text{Cu}(\text{NH}_3)_4(\text{H}_2\text{O})_2]^{2+}$ (or $[\text{Cu}(\text{NH}_3)_4]^{2+}$) [2].	[2]
(d)	$\text{CuO}(\text{s}) + \text{H}_2\text{SO}_4(\text{aq}) \rightarrow \text{CuSO}_4(\text{aq}) + \text{H}_2\text{O}(\text{l})$ [2].	[2]

Question 10: Physical Principles of Suck-Back Prevention and Laboratory Safety in Gas Evolution [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	When heating stops, the air and gas inside the boiling tube cool rapidly [1]; the volume of the gas contracts and pressure inside the tube drops below atmospheric pressure [2]; atmospheric pressure forces the cold limewater solution up the delivery tube into the hot boiling tube [1].	[4]
(b)	The extreme temperature differential between cold liquid and hot borosilicate glass causes sudden uneven contraction (thermal shock), shattering the tube and spraying hot chemicals [2].	[2]
(c)	1. Lift the delivery tube out of the liquid before removing the Bunsen flame [1]; 2. Use a Bunsen valve or safety trap bottle between the reaction vessel and collection liquid [1].	[2]
(d)	Clamped near the mouth at a 45-degree angle pointing away from all personnel [2].	[2]